

Excel Screenshots for Linear Regression Modeling Project

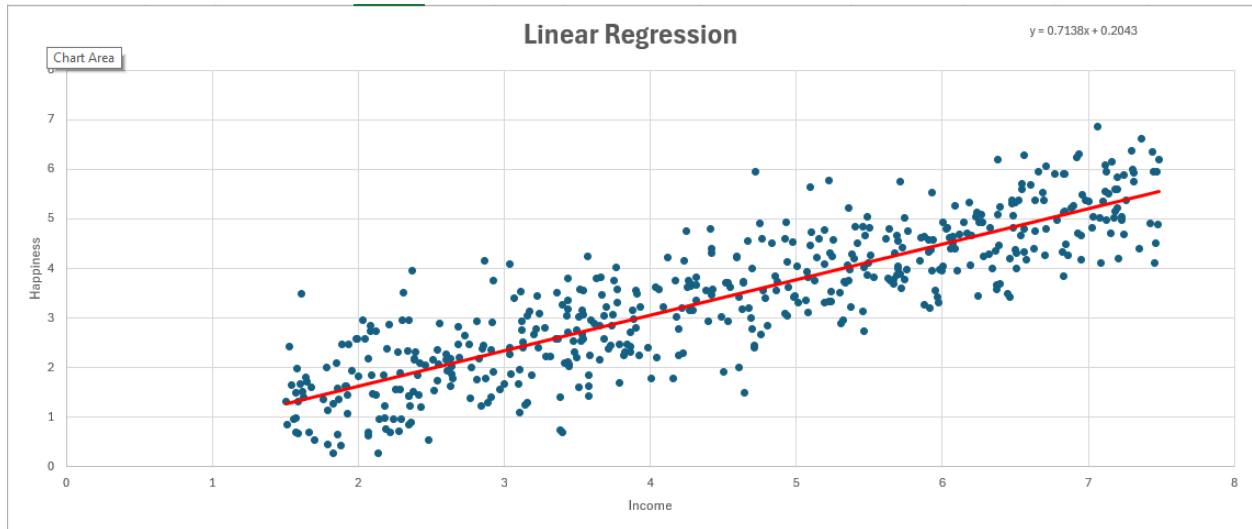
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DSC 101: Introduction to Data Science

Dr. Shawnee Mansfield

May 2, 2025



<i>income</i>		<i>happiness</i>	
Mean	4.466902	Mean	3.392859
Standard Error	0.07786	Standard Error	0.064206
Median	4.42371	Median	3.472536
Mode	#N/A	Mode	#N/A
Standard Deviation	1.737527	Standard Deviation	1.432813
Sample Variance	3.019001	Sample Variance	2.052953
Kurtosis	-1.23136	Kurtosis	-0.76788
Skewness	0.01968	Skewness	-0.01152
Range	5.975246	Range	6.597344
Minimum	1.506275	Minimum	0.266044
Maximum	7.481521	Maximum	6.863388
Sum	2224.517	Sum	1689.644
Count	498	Count	498

SUMMARY OUTPUT								
<i>Regression Statistics</i>								
Multiple R	0.86563373							
R Square	0.749321754							
Adjusted R Square	0.748816355							
Standard Error	0.718100496							
Observations	498							
ANOVA								
	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>			
Regression	1	764.5463593	764.5463593	1482.632007	3.9562E-151			
Residual	496	255.7714878	0.515668322					
Total	497	1020.317847						
	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>	<i>Lower 95%</i>	<i>Upper 95%</i>	<i>Lower 95.0%</i>	<i>Upper 95.0%</i>
Intercept	0.204270396	0.088842247	2.299248426	0.0219055	0.029716856	0.378823937	0.029716856	0.378823937
income	0.713825512	0.018538534	38.50496081	3.9562E-151	0.677401775	0.75024925	0.677401775	0.75024925

RESIDUAL OUTPUT		
<i>Observation</i>	<i>Predicted happiness</i>	<i>Residuals</i>
1	2.961526668	-0.647037686
2	3.758679862	-0.325190104
3	3.719116479	0.880256926
4	2.498771449	0.292342354
5	5.341250916	0.255147357
6	2.866585064	-0.408029191
7	3.541060166	-0.348068357
8	3.41513166	-1.507994827
9	2.432569906	0.509879966
10	3.516359695	0.22158191
11	3.511309436	-0.335903289
12	2.183836242	-0.174789781
13	5.286335851	0.665478247
14	5.534157939	0.426389369
15	1.715968901	-0.270170043
16	2.031068248	0.867514894
17	1.885181873	-0.654014349
18	1.908997967	0.403990088
19	3.598996304	-0.932880269
20	1.627834803	0.956894219
21	5.422988773	0.324455533
22	2.722874483	-0.176349896
23	1.937975305	-0.737189779
24	2.733174213	0.345119169
25	3.935579998	0.382180921
26	4.981185827	0.40029291
27	2.988482519	0.5767418
28	1.839684247	-0.886271251
29	1.903541098	0.265620162
30	2.024246187	0.036548124

31	5.153433842	1.145667417
32	1.528877262	0.0614786
33	2.766206677	-0.515277263
34	5.077184488	0.837063212
35	1.682318846	0.509514856
36	3.933441402	1.834372964
37	1.805462132	-0.832579212
38	5.255418478	-0.244841051
39	3.195679338	-0.956014346
40	1.600860103	0.326718726
41	3.817482526	-0.459410955
42	3.046859166	-0.646771875
43	2.392187436	1.015792603
44	2.833202307	-0.257026
45	2.909261701	-0.436182324
46	4.029458351	-0.276798858
47	3.913393258	0.174237925
48	3.945565226	-0.402361572
49	5.27358764	0.0747653
50	2.648424007	0.407952646
51	1.812804107	-0.254381543
52	4.037753888	-0.812621136
53	4.648266788	0.385964221
54	4.118077339	-0.26065308
55	3.083965368	0.535090121
56	4.851430608	-0.846892869
57	4.50807707	0.294014737
58	5.164724467	-0.505834076
59	5.340271575	-0.108568607
60	2.172528962	0.308077505

61	5.169697396	0.328449797
62	3.537973006	1.012663953
63	4.750120173	-1.180106605
64	4.606204738	0.113460594
65	4.540159551	-0.037051382
66	4.119402721	0.926679053
67	1.338949398	-0.66923345
68	1.403835559	0.202236798
69	4.130273564	0.696329132
70	3.090902892	-0.882662376
71	3.777033643	0.279459457
72	3.676019113	-0.10811388
73	1.279487952	0.028999311
74	2.249140533	1.910468767
75	4.400069868	0.233845189
76	4.832703446	0.236044435
77	3.729166896	-0.688369602
78	4.219849004	-0.415550143
79	5.363990516	-0.329986731
80	4.014285725	-0.310847793
81	2.22141751	-0.032479389
82	4.43823125	1.099816279
83	2.717118369	0.866756792
84	2.51702265	0.57986296
85	2.701507838	-0.500525666
86	5.33390066	-0.182302371
87	3.572931535	2.378054793
88	2.770331745	0.197855357
89	2.51274101	-0.113179684
90	1.285111597	-0.425612545

91	3.061383467	-1.285450915
92	4.628634988	0.032626221
93	1.832257165	-1.105035974
94	1.767452267	-0.9961657
95	2.655655089	0.693133117
96	4.438875837	-0.472660426
97	3.993141602	-1.102694185
98	4.247624303	-0.47436365
99	5.514595133	0.845004871
100	1.72807525	-1.459353152
101	4.845046155	-0.470213838
102	2.810578096	-0.654734773
103	1.836429141	0.057127719
104	3.594127202	1.308864373
105	4.101158623	0.732347804
106	2.654879611	0.517419852
107	5.326967555	-0.29701589
108	4.134875712	0.126137292
109	2.415427533	-0.743036953
110	3.521814433	-2.024790305
111	1.509362109	-0.243873193
112	2.727333949	-0.059868515
113	3.492276225	-1.492950698
114	4.031448117	1.200415196
115	5.114913825	0.096487495
116	3.285878194	0.375778263
117	2.619258872	-1.204224216
118	3.725005751	1.208038341
119	3.72742511	0.403353155
120	2.061325521	0.220941413

121	4.28111602	-0.379945687
122	4.571119776	0.120879075
123	2.896402747	0.681397927
124	5.284723687	0.277821788
125	1.771033058	0.622195062
126	4.452961213	-0.888237473
127	3.004110175	-0.75038866
128	5.259289333	-1.137641586
129	5.165889504	-0.996788746
130	2.817497472	1.006401213
131	1.481369753	-1.022992179
132	2.731456195	-0.154516236
133	3.440323541	-0.49279207
134	3.678701467	0.061294329
135	3.09955056	0.471895951
136	4.226422663	0.581727762
137	4.102925427	-0.085314219
138	2.478643987	-0.638841981
139	3.357712776	0.110861017
140	4.316111803	0.442673689
141	2.857345856	-0.465668337
142	1.762238525	-0.76932116
143	3.267997852	-0.098617613
144	2.638437005	-0.54939465
145	2.760548823	-0.916873048
146	2.709557452	-1.092950017
147	4.958502754	0.990844748
148	4.681231558	0.258996219
149	2.870421267	-0.02918259
150	3.34025459	-0.395863254

151	2.711380422	0.315537753
152	4.658355736	0.439446773
153	2.83220873	0.204611207
154	5.373304357	-0.690487367
155	4.733757196	-0.732938013
156	4.444209433	0.126591694
157	1.95979674	0.082942572
158	2.016736825	-0.268085701
159	4.993042888	1.07097016
160	5.080642555	0.051758914
161	3.832401293	-0.719059203
162	4.509071793	0.312738469
163	4.897384063	-0.71786304
164	2.755694733	-1.124433626
165	4.151660115	-0.329480472
166	1.924148556	-0.066606669
167	3.248585983	0.502503298
168	2.724320601	0.434297993
169	4.589407461	0.337725164
170	3.885966369	0.714148388
171	3.566993496	-1.158613594
172	5.092201489	-0.605497825
173	4.10506991	-1.377354198
174	3.185585735	-0.164861287
175	2.879754955	0.869425772
176	1.827883023	0.483671214
177	1.329520465	-0.341917294
178	1.577766119	-0.116621698
179	4.418871853	0.157984678
180	3.908333162	0.859262477

181	1.546168263	-1.107996696
182	2.020490658	0.330711811
183	2.503783343	0.949319396
184	5.386910688	-0.005512218
185	4.831013944	-0.010922697
186	4.264858059	0.393884624
187	4.939516749	0.440553458
188	4.467765551	-1.151776075
189	2.986575161	-0.186827674
190	4.81647072	-0.609715808
191	4.935531805	-0.53286861
192	2.430663042	0.338455085
193	3.556369029	-0.772124483
194	1.327613119	-0.639522537
195	2.824279089	0.652220051
196	5.33982163	0.496375026
197	1.475221486	0.528704607
198	1.733541303	-0.762208849
199	2.814363381	0.043250983
200	1.696418512	0.155699199
201	2.604934958	0.910244219
202	1.933972298	0.166580196
203	5.280700233	0.805778051
204	2.374258449	1.709562713
205	1.89834397	-0.373561678
206	1.620902589	0.958613926
207	2.080548515	-0.461322388
208	5.298671738	0.207949799
209	2.420640331	-1.32464106
210	1.31687935	-0.348352023

211	5.542576384	-0.664850846
212	2.212904039	-0.453205372
213	4.30486935	0.727341683
214	4.873394761	0.840407533
215	4.88896028	-0.093117412
216	4.108985173	0.551304788
217	1.692689762	1.047705646
218	3.480424335	0.770143729
219	3.826579221	0.1144401
220	5.531914112	-1.028468732
221	4.382937677	0.250291056
222	2.891495063	1.139369875
223	5.245871592	1.617516359
224	4.756604391	0.322871508
225	1.575431242	0.04983715
226	5.461033783	1.157246059
227	4.869690366	-0.212691794
228	5.415841024	0.589146451
229	2.372323887	0.038311679
230	4.989233273	-0.728013301
231	1.580523617	-0.514899807
232	3.219150693	-0.923450551
233	2.290572733	-0.37106163
234	3.364453464	0.216927574
235	1.682290669	-1.053348608
236	3.933345004	-0.589236631
237	5.31659797	0.843836131
238	1.782321942	1.078952502
239	5.348853478	-1.139382905
240	1.763325984	-0.53702785

241	3.355808938	1.437516222
242	3.783969477	-0.470172961
243	2.061670876	0.091027556
244	2.286534158	0.637149716
245	4.661640762	-1.209573529
246	5.100866779	-0.84255525
247	1.896915169	2.064117598
248	4.461566861	-1.033843698
249	5.310745467	-0.589758129
250	1.838857595	-0.266907783
251	4.755784638	1.431160971
252	2.731572691	0.821164273
253	2.445558099	-1.199772155
254	4.816113079	-1.389201427
255	2.089732315	-0.212585272
256	1.633501511	0.184387602
257	4.766634859	0.473274639
258	2.177084079	-0.793039967
259	5.080596315	-1.242618547
260	2.936262318	-0.614564628
261	2.895871481	0.415706748
262	2.459855433	0.597532406
263	3.84436446	0.892681493
264	4.209110501	0.38185996
265	1.529395913	-0.875153766
266	4.033037412	-0.051909672
267	1.871862134	0.464413899
268	3.756160048	0.784476099
269	2.08130797	0.111436894
270	2.093379703	-0.305203361

271	2.959560057	-0.240127877
272	3.146324103	1.066373235
273	4.762717745	-0.285446353
274	2.947294601	-0.478434607
275	3.766651263	-0.314878725
276	2.63170325	-1.944782401
277	2.931166208	-0.684420501
278	1.567319608	0.060375482
279	2.244715013	0.208494048
280	5.523341665	-1.413397012
281	2.598621125	-0.026022922
282	4.992487158	-0.182063099
283	4.719863535	0.107552768
284	1.582796432	0.882964696
285	3.86515682	-0.107777833
286	3.258165749	0.433182439
287	1.331736584	0.650073327
288	2.279848116	-0.869618548
289	4.233611323	-0.479309903
290	2.753466531	1.491749066
291	1.878268006	-0.440391282
292	4.376721159	-0.216834015
293	3.986460888	-0.471569106
294	2.65604683	0.058451208
295	1.536265399	-0.178892263
296	3.841650735	1.796809652
297	1.297000006	1.124464663
298	1.359462402	0.151742705
299	4.978105012	0.544786655
300	5.448921695	-1.052299377

301	4.475692116	-0.487239281
302	3.913366558	-0.595759778
303	3.197346073	-0.417663446
304	1.880109665	-1.033604139
305	3.566024064	-1.108102357
306	1.851206984	1.658397704
307	2.153562849	0.493341688
308	2.975707588	0.002209136
309	2.755763771	-1.331313223
310	3.17348214	-1.383139823
311	1.30686979	0.337766813
312	2.619782173	-1.888145883
313	2.784567728	0.117220863
314	4.846907528	-0.52977437
315	1.523714438	0.573654623
316	3.359601806	1.03199117
317	4.827827623	0.48612928
318	4.887979378	1.393390813
319	5.330185152	0.277781478
320	2.209471062	0.737399373
321	4.263228886	-0.390440038
322	3.63087823	-0.788009879
323	1.926668654	-0.466052324
324	2.29271221	1.460112556
325	2.470078037	0.656525202
326	2.121269959	0.699922686
327	1.720742188	1.014217868
328	2.127332121	0.070259999
329	3.224386757	0.931154108
330	4.023604981	0.054603857

331	3.838769745	0.618193865
332	4.665895576	0.273363384
333	4.005128962	-0.274559004
334	1.666540324	0.908514532
335	1.904163939	0.272158206
336	5.143902232	1.09305308
337	1.888547256	-0.989814453
338	3.449112354	0.180312033
339	5.422693685	0.500652386
340	1.415963285	-0.869773109
341	3.404768341	-0.373271275
342	1.688995582	1.15882126
343	3.655186587	0.861774003
344	3.603039558	0.988657452
345	3.252331424	-0.094566333
346	4.04450562	0.24349406
347	3.241995712	0.376834077
348	3.21409317	-0.013295309
349	4.268596605	-0.268893597
350	2.343263322	-0.681010123
351	2.238286365	0.153263633
352	2.877528121	0.190081332
353	2.660910382	-0.643396794
354	3.259497397	0.381386573
355	1.506613434	-1.240569771
356	2.266685049	-0.971257754
357	1.753255696	0.100091917
358	5.184558685	0.182677418
359	2.119986215	0.346306129
360	1.382907739	0.340685822

361	3.356752694	0.950535722
362	2.974697658	0.181669777
363	2.905786271	-1.215328177
364	4.678000833	0.407522178
365	1.553633864	0.927990006
366	1.462072478	-0.108755137
367	4.550869946	0.724424206
368	4.754375425	-1.084083899
369	4.767387937	-1.070997528
370	2.491593151	0.188294167
371	2.280179855	0.083384814
372	1.881726136	-0.443130031
373	5.204188984	0.158259279
374	2.69095232	-0.365643094
375	2.001300285	-0.471739341
376	4.271970344	-0.236243325
377	3.332199028	0.218138156
378	3.61068792	-0.056087604
379	4.055370778	0.144357748
380	4.099217409	-0.963742255
381	1.480847334	-0.335551522
382	2.426118535	1.113735593
383	3.831195301	-0.008991337
384	3.667262039	0.168315607
385	4.920852562	0.762897999
386	2.43630011	0.081100766
387	4.888937937	-0.564277553
388	1.876817446	1.081879354
389	1.346414784	0.334582349
390	3.950551987	0.302949317

391	2.371060125	-0.110664432	421	1.374502071	0.424216232
392	3.735830854	-0.1203598	422	4.941183742	-0.181169409
393	1.328512468	0.173699158	423	3.491015163	-0.075208683
394	4.302860103	-0.526375473	424	5.223932157	-0.183757576
395	4.423318434	-1.222275999	425	2.418804149	-0.453168217
396	4.626422291	-0.568110675	426	2.377236777	-0.505278923
397	4.12502107	-0.019796554	427	4.617781608	0.705725781
398	3.428794818	0.29330282	428	4.069673327	0.431485009
399	3.282133245	0.540137068	429	2.960133683	-0.525038827
400	3.511953204	0.194707777	430	4.269748541	0.283468006
401	5.503591432	-0.593902963	431	4.747711919	-0.386002649
402	1.994379565	0.161354061	432	1.702656866	-0.223869666
403	4.282060495	1.461675586	433	2.701922677	0.066600452
404	5.074257089	-0.736685637	434	4.483370441	-0.52273407
405	2.608143351	-0.025726712	435	5.087501435	0.82827996
406	4.715784196	-0.431478638	436	4.3122941	-0.332098749
407	1.845100674	1.117067536	437	5.14874054	-0.440537552
408	3.55917598	0.431239914	438	1.363390069	0.053270987
409	5.377188876	0.512959821	439	4.420379767	-0.091962669
410	4.249484706	-0.560945188	440	5.544771228	0.651524318
411	3.122005039	0.095842324	441	2.065822513	-0.119861461
412	2.570285081	-0.349814754	442	1.389685088	-0.691225123
413	5.035681471	0.874603407	443	4.655390307	0.48006501
414	4.063031988	0.79000053	444	2.321470956	-0.76111558
415	2.797086948	1.005886685	445	4.686569999	-0.451578784
416	3.71982513	-0.627660482	446	1.789251321	-1.10034209
417	2.844857673	0.388456074	447	1.682567517	-0.996718642
418	5.127474479	0.138004478	448	2.251820792	-0.460067224
419	5.219264164	-0.407730784	449	3.008277589	0.219194584
420	4.826736884	0.541304289	450	5.083801518	0.075529722

451	2.657806264	1.145164408
452	3.873023171	0.343034421
453	4.490450887	0.441959091
454	2.547878687	-0.314438103
455	4.429495872	-0.051236422
456	2.995469877	0.488455807
457	2.090267777	-0.059970734
458	4.287978646	-0.691580327
459	2.631217476	0.629964628
460	1.354676098	2.126461669
461	3.618987142	-0.225547989
462	3.481792459	0.747870886
463	4.252974303	0.068389994
464	4.524864489	-0.124409344
465	3.284994386	0.072602997
466	4.84117725	0.494540754
467	4.877066841	0.711322495
468	3.550841172	-0.556636888
469	2.436063014	-0.022681261
470	5.045123881	-0.059757291
471	5.286354968	-0.304373844
472	5.413570743	0.970703754
473	4.396919934	-1.129444494
474	1.337336586	-0.024577609
475	4.488513255	-0.448965269
476	4.289655196	0.136339889
477	3.943212087	-0.600462692
478	4.808405933	-1.316095747
479	3.182590756	0.568634707
480	4.860391232	0.526168097

481	5.520926013	0.442496073
482	5.36179668	-0.376541753
483	2.542287943	0.255872785
484	1.654633752	1.312689516
485	2.23163532	-1.002862137
486	4.559722946	-0.597993306
487	4.540280549	0.108863103
488	3.233654894	1.520513501
489	4.000805343	-1.03884868
490	4.524614593	0.108826462
491	2.455706465	-1.161569889
492	1.976257717	-1.427892528
493	2.656333663	-0.541197178
494	3.951289549	0.617415011
495	2.6825288	-0.147526686
496	4.549761454	-0.152310179
497	2.66043468	-0.589770913
498	3.438289292	0.271903686

